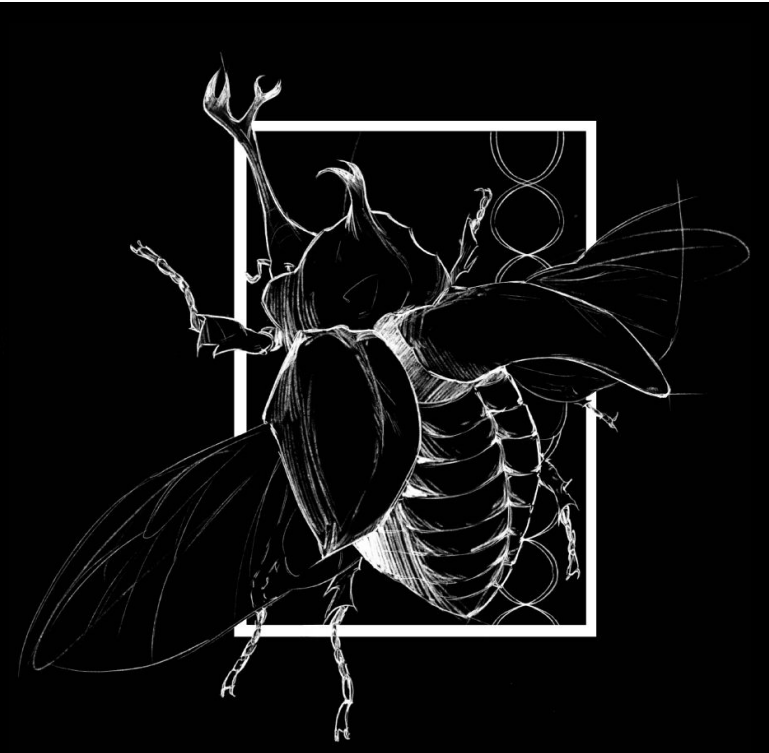




Background

Post-copulatory sexual selection occurs when sperm from multiple males compete to fertilize a female’s ova. The intensity of sperm competition varies across primates depending on mating system structure, with multi-male systems experiencing stronger competition than monogamous or single-male systems. Classic research demonstrates that sperm competition increases relative testes size, reflecting greater sperm production. **However, it remains unclear whether sexual selection also reshapes energetic investment per sperm, the primary question of this study.** Sperm midpiece length reflects mitochondrial volume and ATP production capacity, which may influence sperm performance. If sexual selection drives reproductive escalation, we expect the mating system to shift the adaptive optima of both sperm quantity and energetic investment. This project tests whether sperm competition reshapes male reproductive investment at multiple biological levels across primates.

Study Design & Model Fitting

Sperm midpiece length (μm) and testes mass (g) were collected for 28 primate species from SpermTree and Brindle & Opie (2016). Mating system was coded as monogamy (0), polygyny (1), or polygynandry (3) following Brindle & Opie (2016). Body mass (g) was used as a covariate to control for allometric scaling. Phylogenetically controlled models (PGLS) were used to account for shared ancestry to account for phylogenetic non-independence. Two models were run for each trait: a three-level mating system model (monogamy, polygyny, polygynandry) and a binary model (polygynandry vs. all others). The phylogenetic tree was pruned from a published primate supertree (10k Trees).

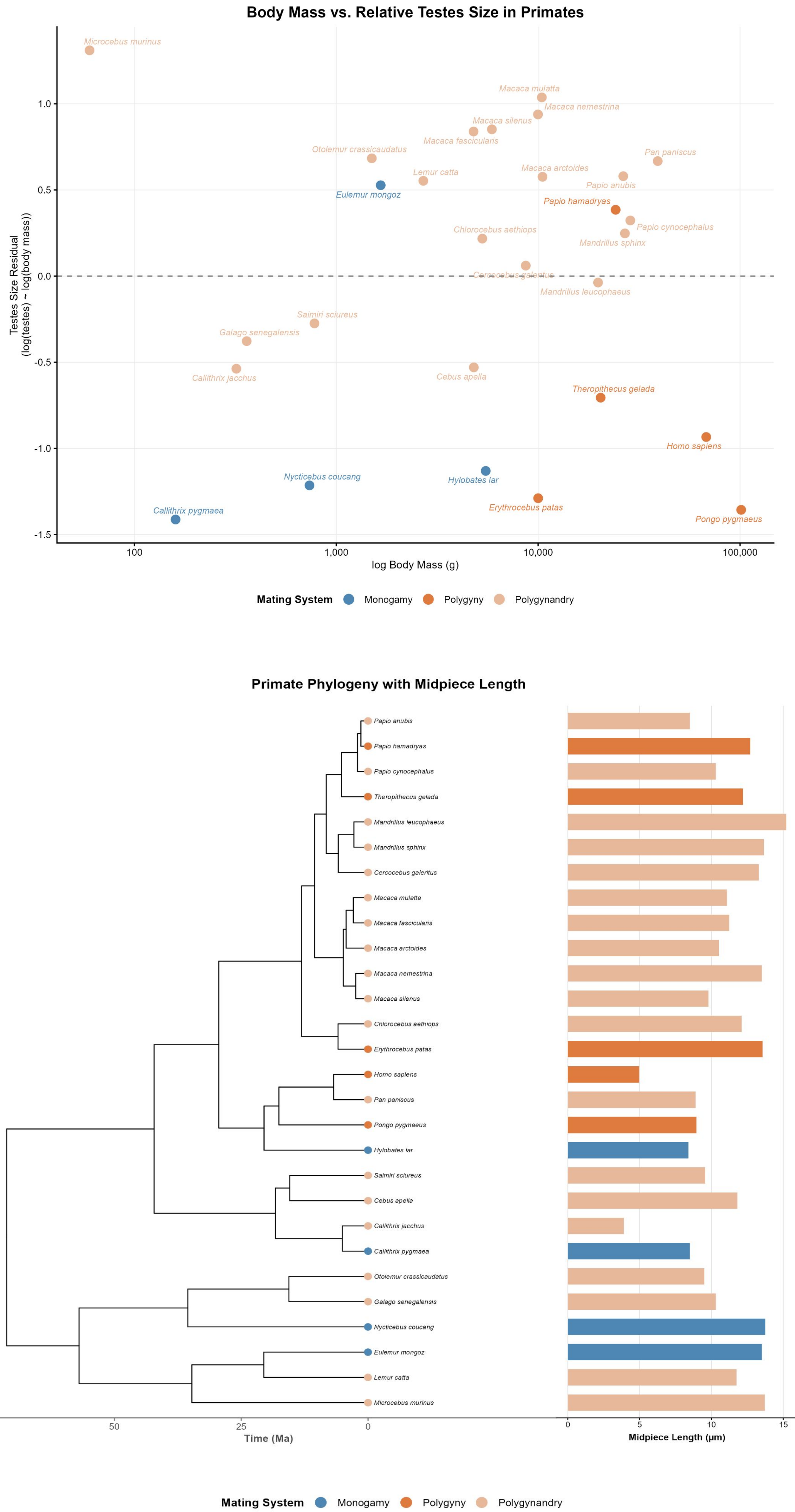
AI Use Statement

I used AI to help narrow down my research topic and identify an appropriate comparative framework. AI assisted in compiling my dataset from two online databases (SpermTree and Brindle & Opie, 2016) and in writing the R scripts used to produce all figures and run the statistical analyses presented here.

Results

Testes mass was significantly predicted by mating system after controlling for body mass and phylogeny ($\lambda = 0.83$, $R^2 = 0.77$). Polygynandrous species had significantly larger testes than all other species combined ($\beta = 1.01$, $p < 0.001$), consistent with established sperm competition literature. Polygynous species did not differ significantly from monogamous species ($p = 0.900$), confirming that sperm competition pressure — present in polygynandrous but not polygynous systems — is the key driver of increased testes investment.

In contrast, midpiece length was not significantly predicted by mating system or relative testes size under any model specification ($\lambda = 0.69$, $R^2 = 0.048$, $p = 0.754$). Neither polygynous ($p = 0.437$) nor polygynandrous species ($p = 0.290$) differed significantly from monogamous species in midpiece length. The near-zero R^2 and moderate phylogenetic signal suggest that midpiece length is phylogenetically conserved across primates rather than shaped by sperm competition pressure.



Discussion

Our results provide partial support for the hypothesis that sperm competition increases male reproductive investment in primates. As predicted, species with higher levels of sperm competition (polyandry) exhibited significantly larger relative testes mass, consistent with extensive prior work showing that increased sperm production is a primary adaptive response to post-copulatory sexual selection. However, contrary to expectations, we found no significant relationship between mating system and sperm midpiece length. Because the midpiece is associated with mitochondrial volume and ATP production, this result suggests that energetic investment per sperm cell may not scale with sperm competition in the same way as sperm quantity.

This decoupling between sperm quantity (testes mass) and sperm quality/energetics (midpiece length) indicates that these traits may evolve under different selective pressures. One possible explanation is that selection in high-competition systems favors producing a greater number of sperm rather than increasing the energetic capacity of individual sperm. Alternatively, sperm performance may be influenced by other factors not captured by midpiece length alone, such as flagellar structure, swimming behavior, or biochemical efficiency.

Additionally, phylogenetic constraints and allometric scaling may limit the extent to which midpiece length can evolve, even under strong sexual selection. This could explain why testes mass shows a clear evolutionary signal while midpiece length does not.

Overall, our findings suggest that sperm competition primarily drives increased investment in sperm production rather than uniform changes across all aspects of sperm morphology. This highlights the importance of considering multiple axes of reproductive investment when studying the evolution of male reproductive strategies.

Conclusions & Acknowledgements

- Sperm competition is strongly associated with increased testes mass in primates
- No relationship was found between mating system and sperm midpiece length
- Male reproductive investment appears to prioritize sperm quantity over energetic investment per sperm
- Different components of sperm biology may evolve independently under sexual selection

Results of PGLS

Response	Predictor	β	p	R^2
Testes Mass	Polygyny vs. Monogamy	-0.05	0.9	0.77
Testes Mass	Polygynandry vs. Monogamy	0.98	0.006**	0.77
Testes Mass	Polygynandry vs. Non-Polygynandry	1.01	<0.001***	0.77
Midpiece Length	Polygyny vs. Monogamy	-0.15	0.437	0.048
Midpiece Length	Polygynandry vs. Monogamy	-0.21	0.29	0.048
Midpiece Length	Polygynandry vs. Non-Polygynandry	-0.11	0.474	0.022